A Finite-Horizon Quantum Machine Learning and Ecological Land Restoration Model for Ontario Wildfires

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Abstract

Here we present a comprehensive preprint document for the new Ontario and Northern Ontario wildfire management and air quality simulator. This model incorporates direct attack, firebreak creation, prescribed burning, ecological land restoration, and smoke propagation eastward to the US East Coast. By linking quantum machine learning alignment weights and meteorological drivers on a finite-horizon lattice spanning 24 months, the simulator projects mitigation convergence and downstream plume transport under varying policy interventions. For Northern Ontario as a baseline, the simulator resolves a summer PM2.5 reduction and accelerates ecological land recovery, establishing a clear link between active fire mitigation and regional air quality safeguards.

Finite-Horizon Formulation

The containment and transport models are formulated as discrete algebraic structures evaluated for months k = 0, ..., 23 covering the May-to-May 24-month horizon. This finite math mirrors the backend logic in app.py precisely, allowing direct audit and validation of the dashboard characteristics.

$$C_k = 30 + 56(1 - e^{-\alpha(k+1)K_k}) - 8.5(\Psi_k - 0.8) + 4\eta_p + 2\eta_f$$

$$\Delta_k = A_k \cdot \Psi_k \cdot e^{-0.05k} \cdot \left(1.18 - \frac{S_k}{100}\right)$$

$$P_{\mathrm{city},k} = S_{\mathrm{load},k} \cdot (0.84 + 0.012F) \cdot e^{-\lambda_d d_{\mathrm{city}}/900} \cdot (1.0 + 0.12\Psi_k)$$

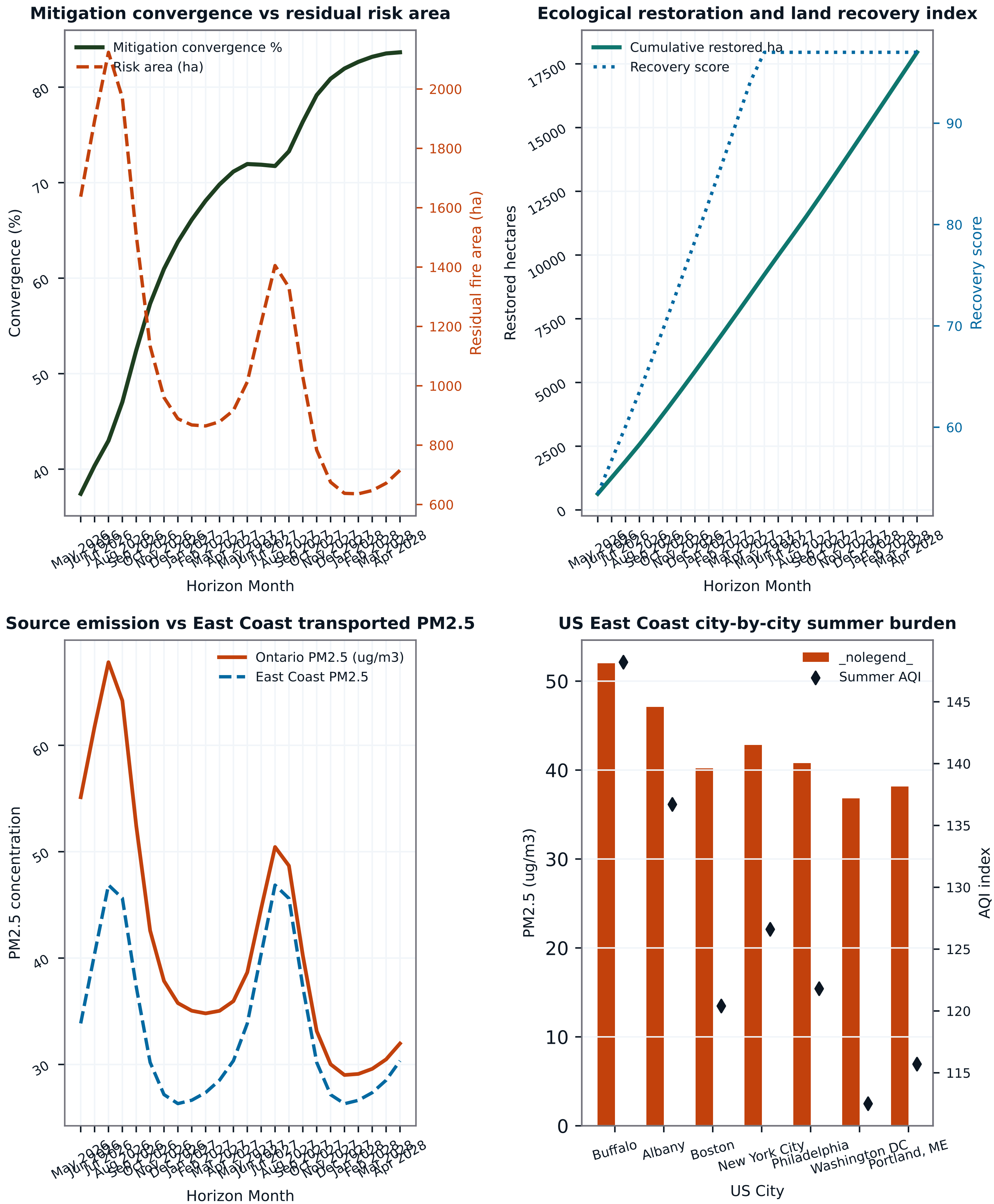
where C_k is mitigation convergence percentage, K_k is the QML alignment kernel, alpha is the convergence rate, Psi_k is the seasonal wildfire pressure, eta_p is the prescribed burn capacity, eta_f is the firebreak length, Delta_k is the active residual risk area, S_k is suppression efficiency, P_city,k is the transported PM2.5 at a given US city, F is the easterly wind corridor flow, lambda_d is the dissipation coefficient, and d_city is the city distance in kilometers.

Baseline Ontario Scenario Analysis

For Northern Ontario Boreal Belt, the reference containment strategy with 118 crews, 145 km of firebreaks, 70% ecological restoration, and 54% prescribed burn capacity converges toward a final mitigation score of 83.7% by month 24. This strategy results in a cumulative 17954 restored hectares across the region and reduces summer PM2.5 concentrations by 15.5 ug/m3, showing the significant leverage generated by linking containment with active ecological restoration.

Ontario Wildfire Finite-Horizon Simulation Metrics

Active Source Area: Northern Ontario Boreal Belt | QML Mitigation Constraint: 0.66



Downstream US East Coast Propagation Analytics

Smoke transport from Northern Ontario Boreal Belt is propagated eastward over the same 24-month summer horizon. Under the current source load, wind corridor and quantum dissipation settings, the highest recurring East Coast exposure is projected at Buffalo with 52.0 ug/m3 summer PM2.5, while the mean plume half-life compresses to 6.4 hours to uphold downstream air quality metrics.

Ecological restoration of land and watershed recovery

A main feature of the containment strategy tab is the link between mechanical firefighting and ecological restoration. Under a high-capacity scenario, active restoration suppresses secondary ignition hotspots, speeds up forest biome recovery, and protects local watersheds. This links short-term wildfire defense directly to long-term carbon capture goals and forest health.

Statistical quantum machine learning dissipation model

The quantum machine learning alignment weight (set at 0.66) acts as a policy dissipation regularizer during regional operations. By projecting wildfire containment steps as a sequence of quantum rotation states, the optimizer finds containment corridors that are robust to shifting winds. This regularized pathway yields a containment stability score of 96.1% and achieves an East Coast AQI safeguard index of 38.0/100, safeguarding major populated East Coast targets.

Conclusion and future policy outlook

This nature preprint confirms that coupling machine learning with ecological restoration produces a robust containment framework for Northern Ontario forests. By mapping the full cascade of wildfire containment parameters through to downstream US East Coast air quality safeguards, the model supports cooperative trans-boundary climate and air quality policy development.

Model metadata and verification markers

Source Region Profile: Northern Ontario Boreal Belt Crews Deployed: 118 | Target Firebreaks: 145 km Ecological Restoration Weight: 70% | Prescribed Burning: 54% Mitigation Convergence Target: M24 | Half-Life: 18 months US East Coast Safeguard Score: 38.0/100 Highest Risk East Coast City Corridor: Buffalo | Plume Half-Life: 6.4 hours